[3]

DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
THIRD SESSIONAL
SUBJECT: Machine Learning (CE622)

Examination	: B.Tech Semester VI		
Date	: 20/03/2023	Seat No.	: CE 137
Time	: 2:30 PM to 3:45 PM	Day	: Monday
		Max. Marks	: 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed. [12]

CO1 N (a) Which Machine Learning Algorithm you would use to automatically identify the category of a product based on the description of the product?Justify your answer. [2]

CO2 R (b) Define expectation of a random variable. Give an example. [2]

CO1 U (c) What is dimensionality reduction? Why is it required? [2]

CO3 U (d) What is the influence of choosing very large and small values of hyperparameter 'c' in Linear SVM? [2]

CO2 U (e) What are the advantages of an SVM classifier over a Logistic Regression classifier? Explain in brief. [2]

CO4 E (f) Consider a single layer perceptron with sigmoid activation function. What is the output of the perceptron if the Input is given as [x0, x1, x2] =[1, 0.7, 0.8] , the weights are [w0, w1, w2]=[0.9, 0.2, 0.3] and bias=0. Show equations and steps clearly. [2]

Q.2 Attempt *Any TWO* from the following questions. [12]

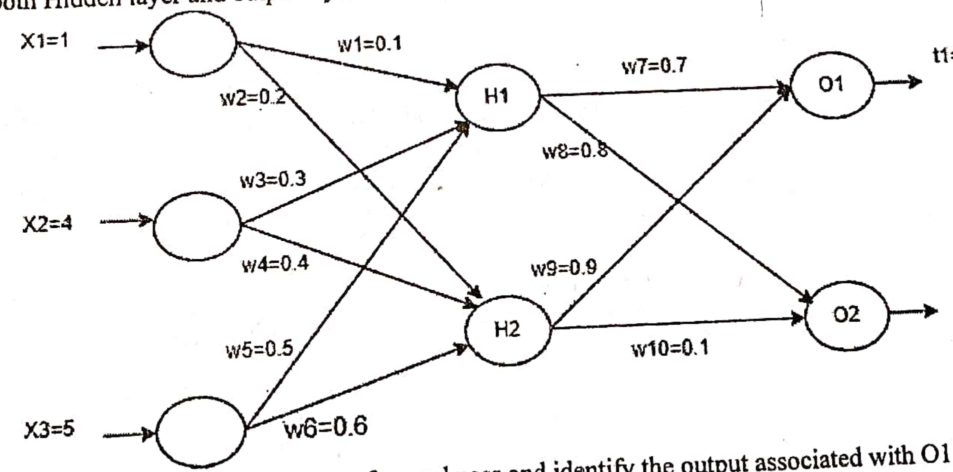
CO3 N (a) i) For Hard Margin SVM, write the primal constrained optimization and dual lagrangian optimization equations. Write and explain the constraints in both the cases [3]

CO2 ii) Compare Hard Margin SVM and Soft Margin SVM. [3]

CO2 U (b) i) What is Kernel Trick in SVM? Explain its importance with an example. [2]

CO4 E ii) Consider the following data points : **X1 = [2 3] , Y1 = 1** and **X2=[4 5] , Y2 = -1**. Using the given data points find the values of α's associated with each data point, weight vector, bias and the decision boundary for Linear SVM. [4]

CO4 C (c) Consider the following Multi Layer Perceptron Network. Sigmoid Activation function is used in both Hidden layer and output layer. Learning rate value = 0.5. [6]



i) For the given network perform forward pass and identify the output associated with O1 and O2.
ii) Find the error associated with each neuron of hidden layer and output layer.
iii) Find the updated weight value for w3.

Q.3 Answer the following questions. [12]

CO4 N (a) Explain the working of the principal component analysis technique for the following data set. [6]

X1	1	3	5	7	9	11	13	15
X2	16	14	12	10	8	6	4	2

CO4 N (b) Describe the working of Expectation Maximization algorithm to find out the missing data in the given data points. The joint distribution is defined as P(X1, X2, X3) =P(X1|X2)*P(X2|X3)*P(X3) [6]

X1	1	0	1	0	?	0	1	1
X2	0	0	?	1	1	0	1	0
X3	1	?	0	1	0	1	0	1

OR

Q.3 Answer the following questions. [12]

CO4 C (a) Perform singular value decomposition and divide the following matrix A into three sub matrices. A = [(2, 5), (3,6)] [6]

CO2 N (b) Describe the Latent Dirichlet Allocation technique for topic modeling. List out two applications of it. [6]